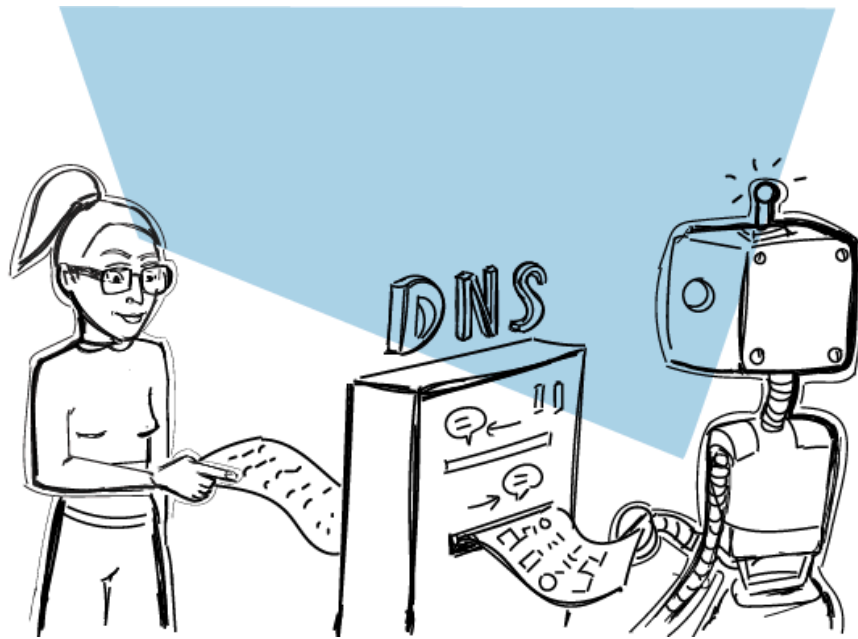
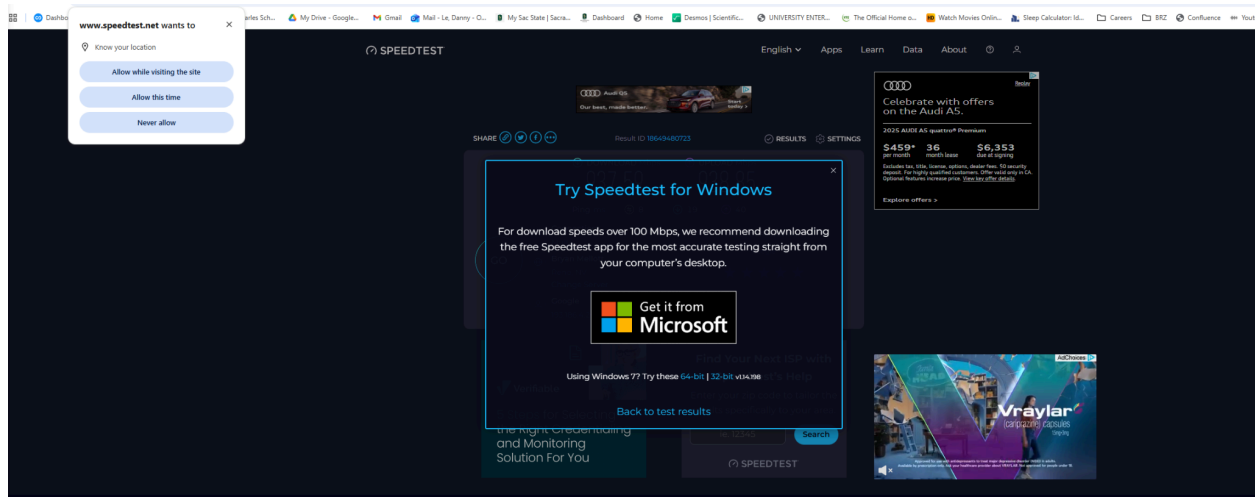
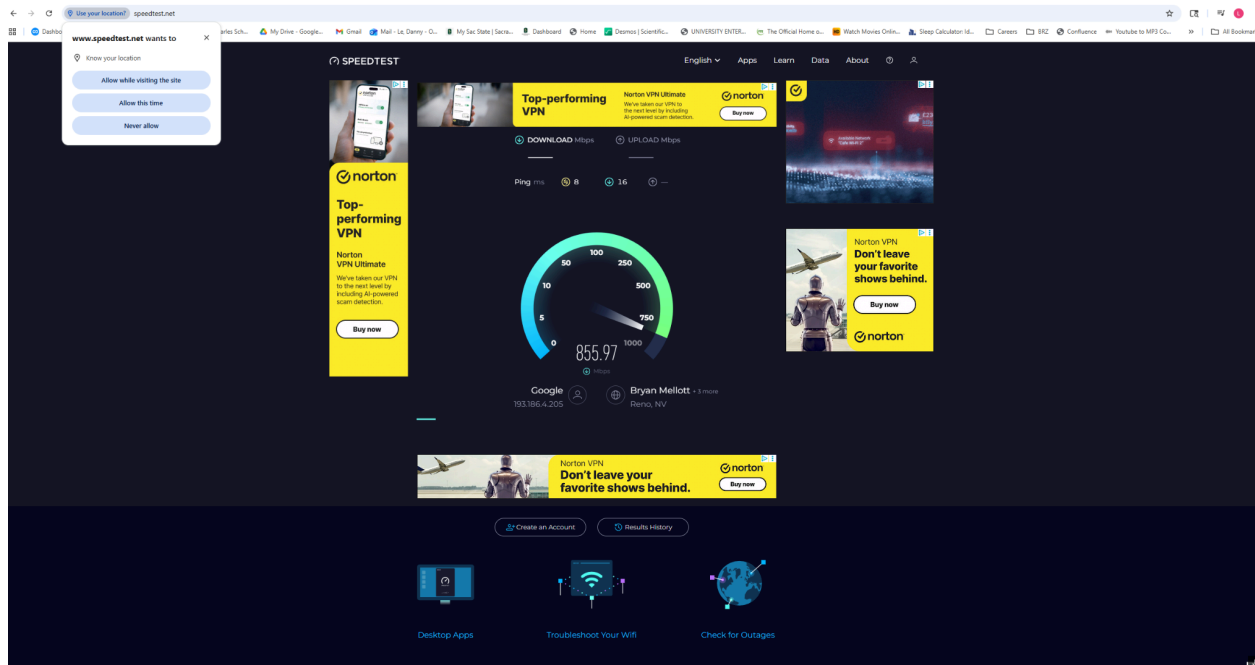


Pi-hole_DNS_Traffic_Filtering_Notes



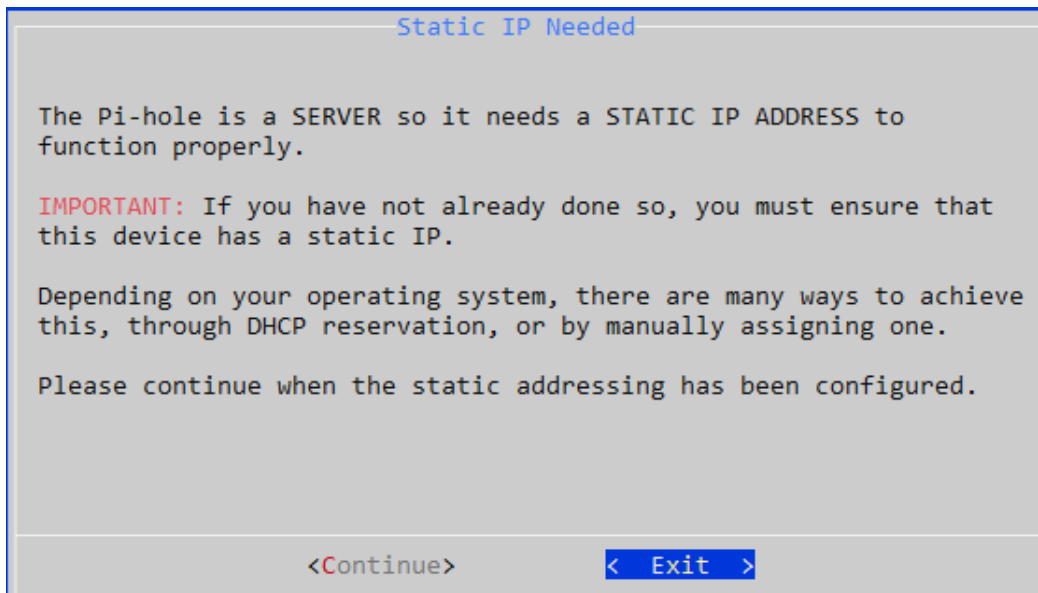
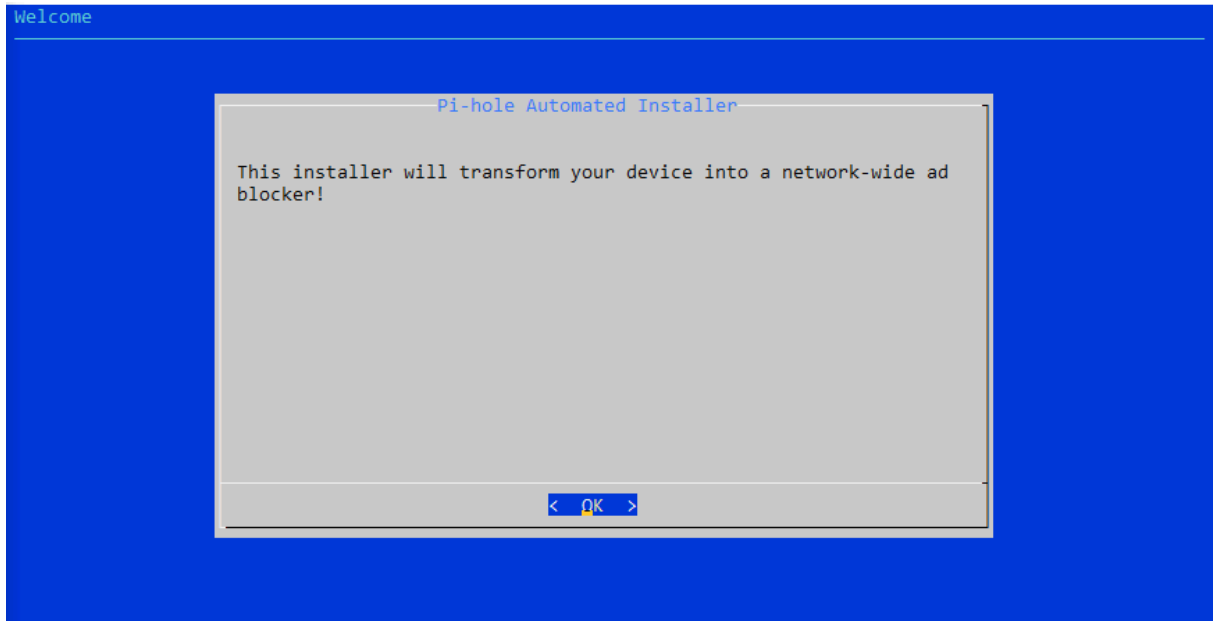
Normally an edge device will request for a URL's associated IP and the DNS server will return the IP and the edge device will be able to connect to the URL. In this project, I will use Pi-Hole to make my Pi into another DNS server that will essentially have a list of IPs that are correlated to ads/malware. If these IPs are requested from an edge device, the DNS will provide an invalid IP address instead of the requested URL's IP. (this is the ad/malware blocking). If the DNS is unfamiliar with the URL, it will send the request to the next upstream DNS server. Likely will be my ISPs provided DNS. This DNS server will indirectly be a backup DNS in case my Pi goes down thus it's DNS along with it. So yes, if ads/malware are not in the list, it can get through and eventually to an edge device. Not 100% ad blocking, but a good bit of it. There are different lists from the community or Pi-hole devs.

Before on [speedtest.net](https://www.speedtest.net)



After on speedtest.net

Installing Pi-Hole using command
`curl -sSL https://install.pi-hole.net | bash`



Ran into this issue

Static IP - an IP address that a device keeps forever rather than getting a new one assigned every now and then

DHCP = Dynamic Host Configuration Protocol

basically the system in which automatically assigns IP addresses and network setting to edge devices

router = DHCP server

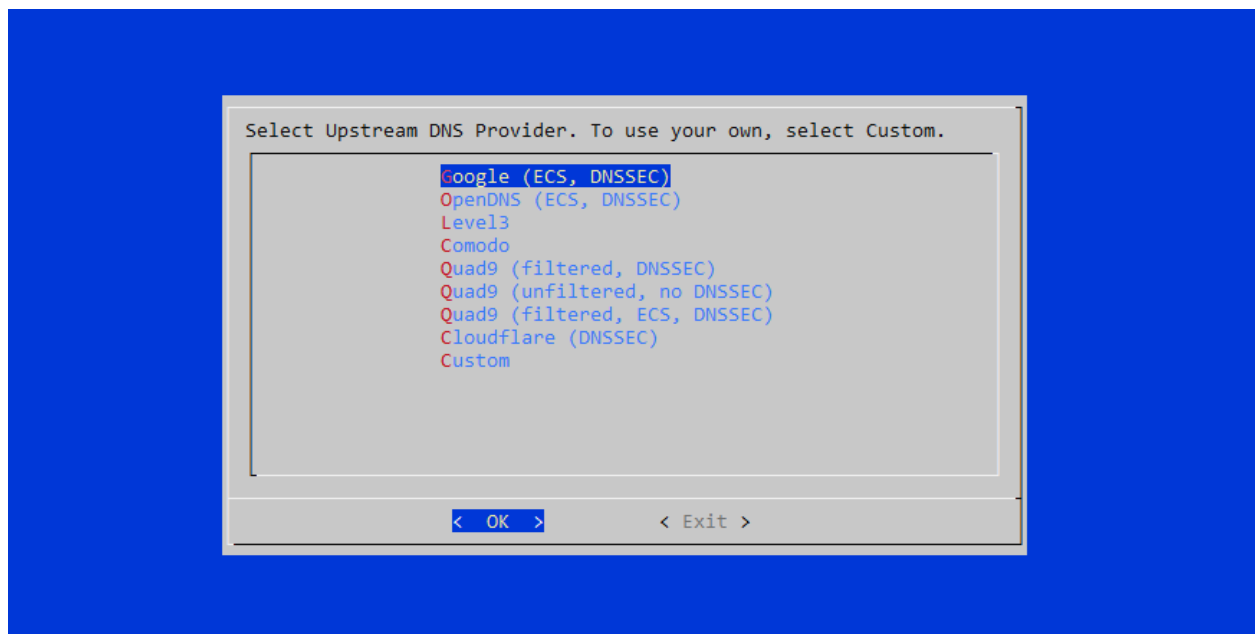
edge device = DHCP client

DHCP Reservation - reservation of an IP address for a device so it never changes.
Makes a device's IP static

IP Allocation

IPv4 Address / Name	MAC Address	Status	Allocation	Action
192.168.1.10	b8:27:eb:xx:xx:xx	on	Fixed Allocation	<button>Allocate</button>

I went to my router's portal by entering its IP and made my Pi's IP into a static IP



Turns out Pi-hole makes setting up an Upstream DNS easy. I will be going with Google. Upstream meaning the DNS server that will be used to obtain IPs of domains not listed in the ad blocking DNS server has since that DNS server only has a list of known ad or malware providing domains

```
Installation Complete!
Configure your devices to use the Pi-hole as their DNS server
using:

IPv4: 192.168.1.1
IPv6: 2608:8b58:4001::1
If you have not done so already, the above IP should be set to
static.
View the web interface at http://pi-hole.local/admin or
http://192.168.1.1/admin
Your Admin Webpage login password is 12345678

To allow your user to use all CLI functions without
authentication,
refer to https://docs.pi-hole.net/main/post-install/

< OK >
```

Finishing install **pi.hole:80/admin**



This site can't be reached

Check if there is a typo in pi.hole.

If spelling is correct, [try running Windows Network Diagnostics](#).

DNS_PROBE_FINISHED_NXDOMAIN

Reload

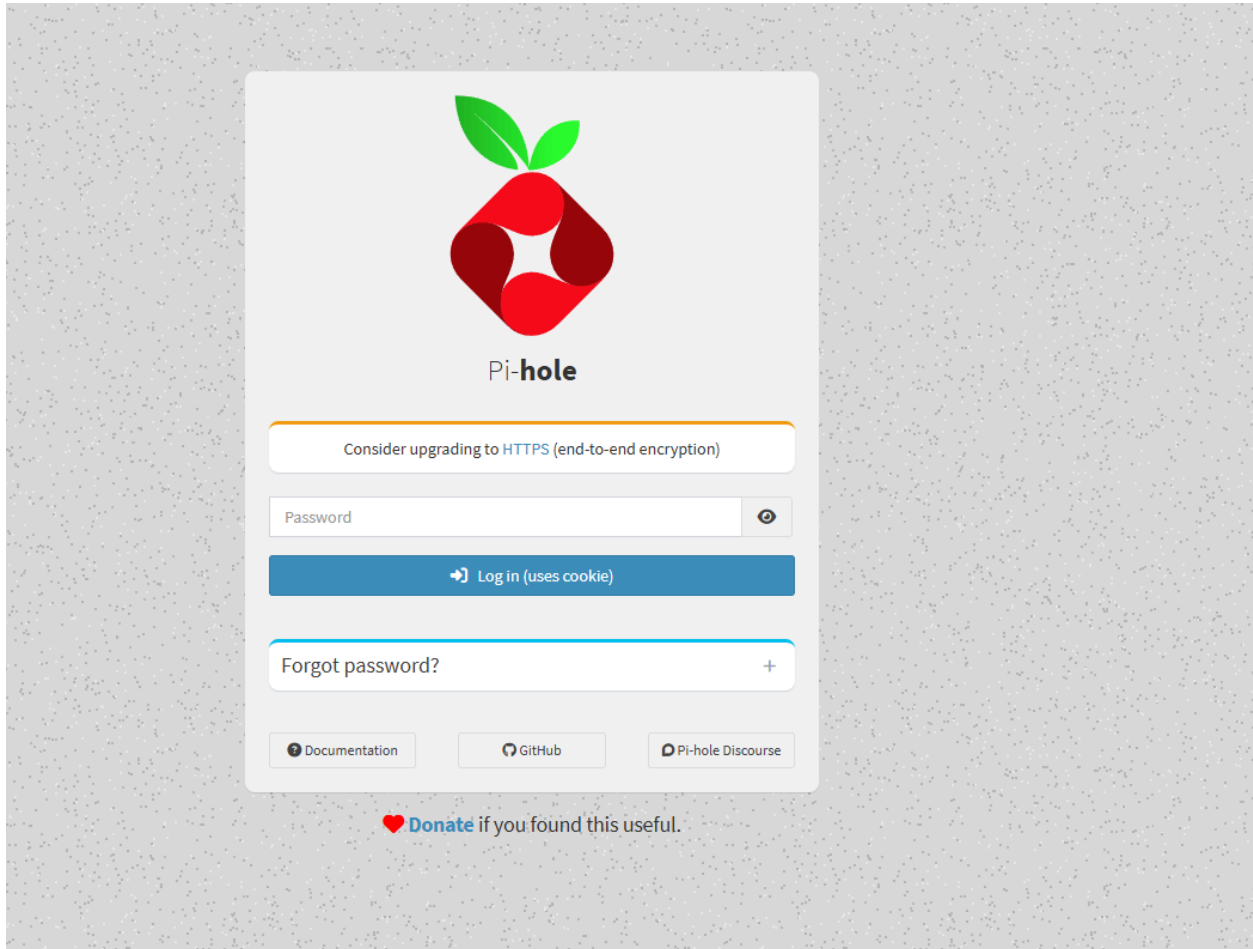
Pi-hole struggling to run due to my NAS service using port 80 already. Pi-hole Admin dashboard uses port 80 as default as well.

```
danny@dannypi:~ $ sudo nano /etc/pihole/pihole-FTL.conf
```

```
GNU nano 7.2  
WEB_PORT=8081
```

```
/etc/pihole/pihole-FTL.conf *
```

Changing port to 8081 for the Admin dashboard following with reboot.



Pi-hole admin portal now runs on port 8081 and runs in parallel with OpenMediaVault

I have to look into how others have set these two services up at the same time. Maybe docker. . . Both require the same port